

Electronics-Retailer-Dashboard

1. Project Title

Electronics Retailer Analysis Dashboard Using Power BI

2. Project Objectives

The primary objective of this project is to analyse store sales performance and provide actionable business insights through an interactive Power BI dashboard. The dashboard helps stakeholders monitor sales trends, identify top-performing and underperforming products, evaluate promotional effectiveness, and understand customer purchasing behaviour across different cities.

The project aims to:

- Analyse overall sales and profit performance.
- Identify the top and bottom 5 products based on sales, profit, and quantity sold.
- Track sales trends over daily, monthly, quarterly, and yearly periods.
- Examine the relationship between sales and profit.
- Evaluate discount strategies through average discount analysis.
- Measure business performance using total order counts.
- Compare sales performance across different cities.

3. Dataset Information

The dataset contains transactional retail sales data collected from a store business. It includes information related to products, customers, promotions, sales transactions, discounts, and profitability.

Key Data Fields

Fact Table

- Customer ID
- Product ID
- Date
- Net Sales
- Profit
- Discount Percentage

- Discount Value
- Price Per Unit

Dimension Tables

- Product Details
- Customer Information
- Promotion Information

The dataset was cleaned and transformed using Power Query to ensure data quality and consistency before analysis.

4. Data Model

A **Star Schema** data model was implemented to improve performance and simplify reporting.

Tables Used

Fact Table

- Stores transactional sales records.

Dim Product

- Contains product-related information such as Product ID, Product Name, Product Line, and Price.

Dim Customers

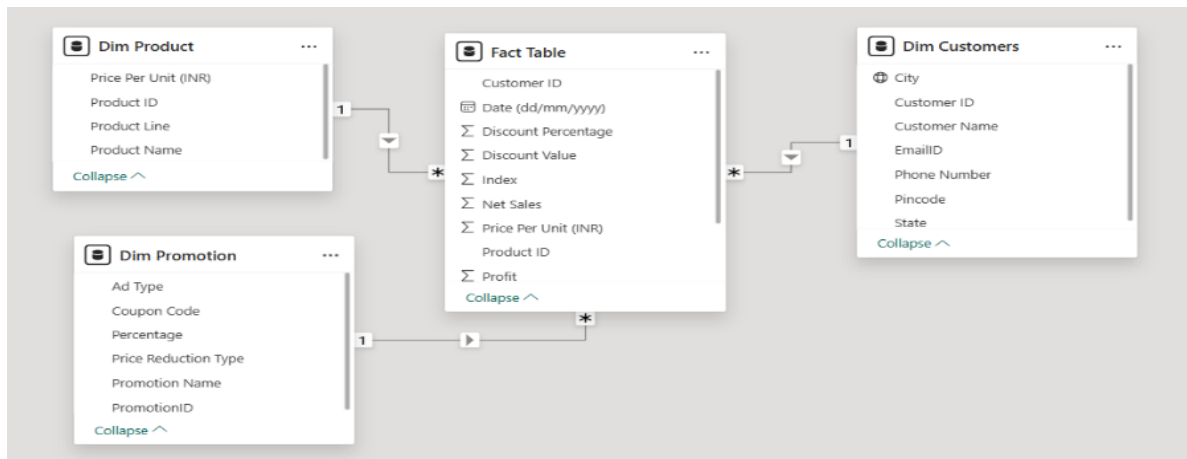
- Contains customer details including Customer ID, Customer Name, City, State, and Contact Information.

Dim Promotion

- Contains promotion details such as Promotion ID, Promotion Name, Coupon Code, Ad Type, and Discount Percentage.

Relationships

One-to-many relationships were established between dimension tables and the fact table, enabling efficient filtering and accurate aggregations across reports.



5. DAX Measures

The following DAX measures were created to support business analysis:

Total Sales = SUM('Fact Table'[Net Sales])

Total Profit = SUM('Fact Table'[Profit])

Total Orders = COUNTROWS('Fact Table')

Average Discount =

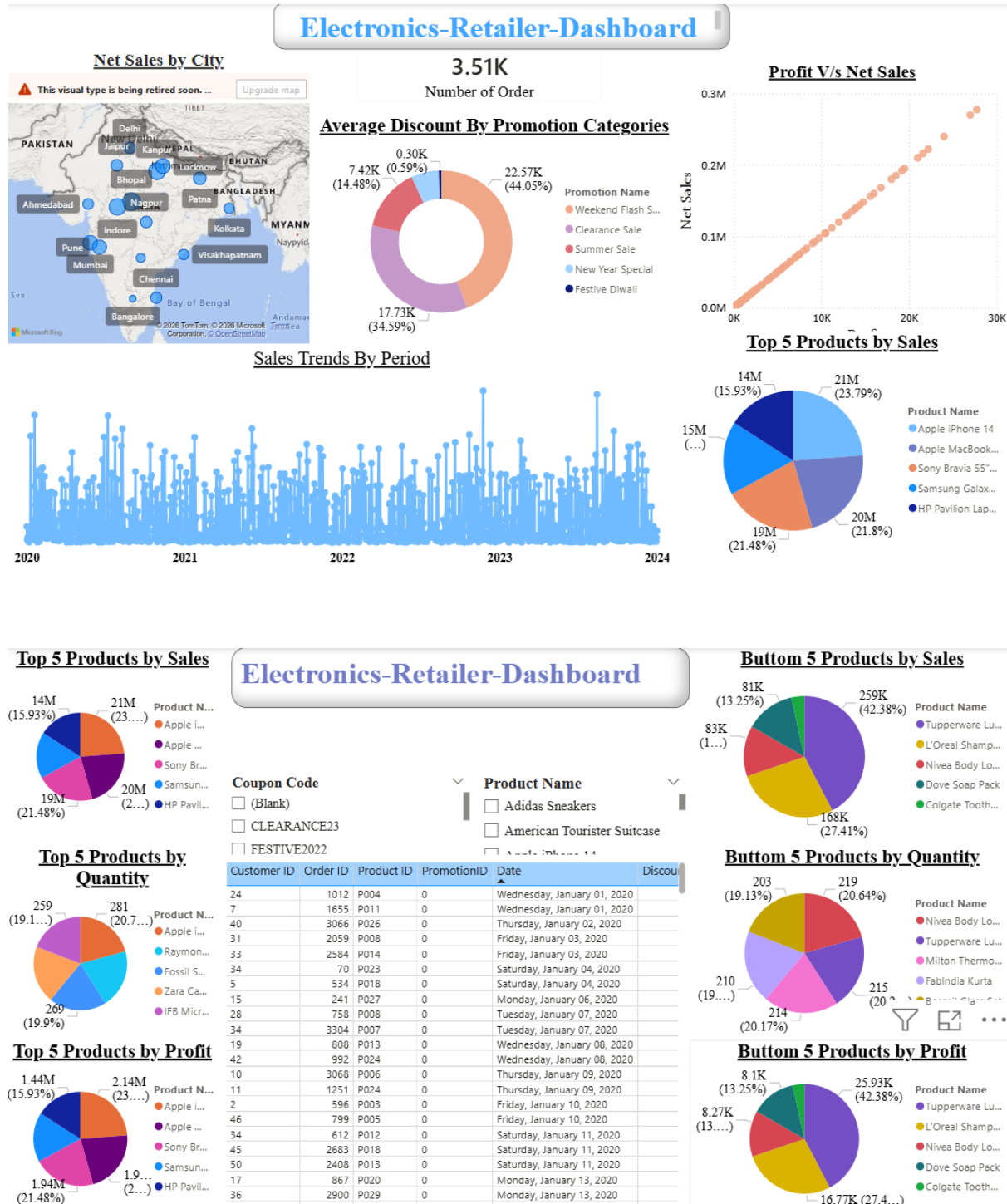
AVERAGE('Fact Table'[Discount Percentage])

Profit Margin % =

DIVIDE([Total Profit], [Total Sales], 0) * 100

These measures were used to build KPIs, charts, and analytical visualizations throughout the dashboard.

6. Dashboard Screenshots



7. Tools Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Data Modeling (Star Schema)
- Microsoft Excel (Dataset Source)

Visualizations Used

- KPI Cards
- Bar Charts
- Column Charts
- Line Charts
- Scatter Charts
- Slicers and Filters
- Tables and Matrices

8. Conclusion

The Store Sales Analysis Dashboard successfully transforms raw sales data into meaningful business insights. Through interactive visualizations and KPI tracking, the dashboard enables users to monitor sales performance, evaluate profitability, analyse product performance, and assess the impact of promotional campaigns.

The project demonstrates practical skills in data cleaning, data modeling, DAX calculations, business intelligence reporting, and data visualization using Power BI. The insights generated from the dashboard can support better business decision-making, improve sales strategies, and enhance overall operational performance.